Social Media Dataset Record Manager

Program purpose
This console application loads social-media usage records from a text file and stores them in a dynamically allocated array. Users can display records, search by participant ID, calculate the average daily usage, and inspect a record through a pointer. Admin users may also add, delete, and save records, while Regular Users have read-only access.

How the program works

- At startup, **RecordManager** allocates an array with **new[]** and loads pipe-delimited records from **records.txt**.
- Each file row becomes a **DatasetRecord** object. The insertion function places it in participant-ID order.
- A **User pointer** refers to either an **Admin** object or a **RegularUser** object. Virtual functions determine which actions are allowed.
- Binary search repeatedly halves the sorted search range until it finds the requested participant ID or returns-1.
- Admin changes can be written back to the text file. When the program exits, the RecordManager destructor releases the dynamic array with **delete[]**.

Project files

main.cpp - menus and input validation
DatasetRecord.h/.cpp - individual record class
RecordManager.h/.cpp - dynamic array, search, sorting, and file I/O
User.h/.cpp - User, Admin, and RegularUser classes
records.txt - persistent program data

Compile and run

```
g++ -std=c++17 -Wall -Wextra -Wpedantic main.cpp DatasetRecord.cpp RecordManager.cpp
User.cpp -o record_manager.exe
.\record_manager.exe records.txt
```

Required concepts demonstrated

Requirement	Implementation
Data types and sizes	Uses char, bool, int, double, string, objects, and pointers. Menu option 7 prints sizeof results for each.
File types	C++ source/header files and the pipe-delimited records.txt data file.
Pointers	DatasetRecord* addresses the first array element; User* provides polymorphic role behavior.
Arrays	RecordManager owns a dynamically allocated DatasetRecord array.
Dynamic memory	The constructor uses new DatasetRecord[capacity], and the destructor uses delete[].
Binary search	binarySearch() searches the participant-ID-sorted array in O(log n) time.
Strings	Participant IDs, usernames, role names, platforms, and filenames use std::string.
File I/O	ifstream loads records.txt and ofstream saves Admin changes.
Two or more classes	DatasetRecord, RecordManager, User, Admin, and RegularUser are implemented.

Access-control summary

Action	Admin	Regular User
Display / search / calculate / pointer / sizes	Allowed	Allowed
Add / delete / save	Allowed	Denied

```
PS C:\Users\Fredd\Downloads\Social_Media_Record_Manager_Submission> g++ -std=c++17 -Wall -Wextra -Wpedantic main.cpp DatasetRecord.cpp RecordManager.cpp User.cpp -o record_manager.exe
PS C:\Users\Fredd\Downloads\Social_Media_Record_Manager_Submission> .\record_manager.exe records.txt
Loaded 8 records from records.txt.
Enter your username: user

Choose your role:
1. Admin
2. Regular User
Choose an option: Admin
Invalid choice. Enter a number from 1 to 2: 1

Welcome, user!
Role: Admin

=== SOCIAL MEDIA RECORD MANAGER ===
1. Display All Records
2. Binary Search by Participant ID
3. Calculate Average Daily Hours
4. Add Record (Admin only)
5. Delete Record (Admin only)
6. Demonstrate Pointer Access
7. Display Data Type Sizes
8. Save Records to File (Admin only)
9. Exit
Choose an option: 1

=== SOCIAL MEDIA DATASET RECORDS ===
Participant ID  Daily Hours     Primary Platform
-----------------------------------------------------
P0001           7.5             Instagram
P0002           8.0             TikTok
P0003           6.5             TikTok
P0004           6.0             YouTube
P0005           7.0             Instagram
P0006           6.5             YouTube
P0007           5.5             Instagram
P0008           5.0             YouTube
```

```
=== SOCIAL MEDIA RECORD MANAGER ===
1. Display All Records
2. Binary Search by Participant ID
3. Calculate Average Daily Hours
4. Add Record (Admin only)
5. Delete Record (Admin only)
6. Demonstrate Pointer Access
7. Display Data Type Sizes
8. Save Records to File (Admin only)
9. Exit
Choose an option: 2
Enter the participant ID to find: P0009
Record not found.

=== SOCIAL MEDIA RECORD MANAGER ===
1. Display All Records
2. Binary Search by Participant ID
3. Calculate Average Daily Hours
4. Add Record (Admin only)
5. Delete Record (Admin only)
6. Demonstrate Pointer Access
7. Display Data Type Sizes
8. Save Records to File (Admin only)
9. Exit
Choose an option: 2
Enter the participant ID to find: P0003
Record found at sorted array index 2:
P0003           6.5             TikTok

=== SOCIAL MEDIA RECORD MANAGER ===
1. Display All Records
2. Binary Search by Participant ID
3. Calculate Average Daily Hours
4. Add Record (Admin only)
5. Delete Record (Admin only)
6. Demonstrate Pointer Access
7. Display Data Type Sizes
8. Save Records to File (Admin only)
9. Exit
Choose an option: 3
Average daily social media use: 6.50 hours

=== SOCIAL MEDIA RECORD MANAGER ===
1. Display All Records
2. Binary Search by Participant ID
3. Calculate Average Daily Hours
4. Add Record (Admin only)
5. Delete Record (Admin only)
6. Demonstrate Pointer Access
7. Display Data Type Sizes
```

```
=== SOCIAL MEDIA RECORD MANAGER ===
1. Display All Records
2. Binary Search by Participant ID
3. Calculate Average Daily Hours
4. Add Record (Admin only)
5. Delete Record (Admin only)
6. Demonstrate Pointer Access
7. Display Data Type Sizes
8. Save Records to File (Admin only)
9. Exit
Choose an option: 4
Enter new participant ID: P0009
Enter average daily hours: 12
Enter primary platform: Instagram
Record added in sorted order.

=== SOCIAL MEDIA RECORD MANAGER ===
1. Display All Records
2. Binary Search by Participant ID
3. Calculate Average Daily Hours
4. Add Record (Admin only)
5. Delete Record (Admin only)
6. Demonstrate Pointer Access
7. Display Data Type Sizes
8. Save Records to File (Admin only)
9. Exit
Choose an option: 6
Pointer address: 0x1fa0363ede8
First record accessed through the pointer:
P0001           7.5             Instagram

=== SOCIAL MEDIA RECORD MANAGER ===
1. Display All Records
2. Binary Search by Participant ID
3. Calculate Average Daily Hours
4. Add Record (Admin only)
5. Delete Record (Admin only)
6. Demonstrate Pointer Access
7. Display Data Type Sizes
8. Save Records to File (Admin only)
9. Exit
Choose an option: 7

=== C++ DATA TYPE SIZES ON THIS COMPUTER ===
char: 1 byte(s)
bool: 1 byte(s)
int: 4 byte(s)
double: 8 byte(s)
string object: 32 byte(s)
DatasetRecord object: 72 byte(s)
```

```
=== SOCIAL MEDIA RECORD MANAGER ===
1. Display All Records
2. Binary Search by Participant ID
3. Calculate Average Daily Hours
4. Add Record (Admin only)
5. Delete Record (Admin only)
6. Demonstrate Pointer Access
7. Display Data Type Sizes
8. Save Records to File (Admin only)
9. Exit
Choose an option: 8
Records saved to records.txt.

=== SOCIAL MEDIA RECORD MANAGER ===
1. Display All Records
2. Binary Search by Participant ID
3. Calculate Average Daily Hours
4. Add Record (Admin only)
5. Delete Record (Admin only)
6. Demonstrate Pointer Access
7. Display Data Type Sizes
8. Save Records to File (Admin only)
9. Exit
Choose an option: 9
Goodbye! Dynamic memory has been released.
PS C:\Users\Fredd\Downloads\Social_Media_Record_Manager_Submission>
PS C:\Users\Fredd\Downloads\Social_Media_Record_Manager_Submission>
```